

# MEGARA 2.1

Mt John Echelle General Algorithms for Reduction and Analysis

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Manual for MEGARA 2.1

## **Abstract**

MEGARA is a MATLAB pipeline for reducing and post-processing HERCULES echelle spectra. Version 2.1 handles frames obtained with CCD486 (data prior to April 2026). MEGARA is under active development; users should retain their original data and use the most recent release.

## Authorship and development assistance

MEGARA was developed and is maintained by E. Brunsden. It was originally built from reduction and line-profile analysis code developed by Duncan J. Wright during his PhD research at the University of Canterbury [1]. Many MEGARA 2.1 code and manual updates were implemented with assistance from artificial-intelligence tools, including OpenAI ChatGPT and Codex. Scientific decisions, review, validation and responsibility remain with the human developers and contributors.

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## 1 Things every user should know

- Information about HERCULES is provided by Hearnshaw et al. [6].
- Information on getting started with MATLAB is available from [MathWorks](#).
- MEGARA includes a worked tutorial; see Appendix A.
- Other code supplied with MEGARA that users may find useful is described in Section 15.
- For common problems and errors, see Section 16.
- For support, contact [emily.brunsdan@york.ac.uk](mailto:emily.brunsdan@york.ac.uk).
- Run MEGARA using `User_codes/Run_megara.m`. Review all user options before starting a reduction.
- Do not edit routine files in `Reduction_code` unless you are developing or debugging MEGARA.
- Keep the MEGARA program, raw observations and reduced products in separate directories. Never place raw or reduced data inside the `Version_2_1` program folder.
- The `J*.mat` wavelength arrays are not barycentrically shifted. The correction is stored in each reduced frame and is applied when `final_data` is built if `options.apply_barycentric_correction` is true.
- Automatic continuum fitting requires SUPPNet. Synthetic spectra and LSD masks require the bundled SynSpec installation. Manual and template continuum fits can be run without this installation.

## 2 Summary of changes since MEGARA 1.6

MEGARA 2.1 contains substantial changes to its interface, reduction methods and output structure. Users upgrading from 1.6 should install 2.1 in a new program directory and should rebuild final products before using reduction products (`J*.mat` files) before using the new continuum, cosmic-ray, LSD or radial-velocity results.

- `Run_megara.m` now passes one `options` structure. Options use `true` and `false` rather than numeric switches.
- Raw and reduced data are located using separate pointer files and must remain outside the MEGARA program directory.
- The barycentric correction is calculated entirely in MATLAB. The relevant HRSP C code and runtime catalogues have been translated into MATLAB code and data files, so the HRSP executable is no longer required.
- Automatic continuum fitting is a two-stage procedure: a synthetic-guided fit places the orders on a common scale, then SUPPNet refines the complete merged spectrum.
- A separate template-continuum mode is available for the standard wavelength range and does not run SUPPNet. This uses a range of main-sequence spectral standards.
- The standard wavelength range is 3869–6866 Å. An optional extended mode covers approximately 3750–8950 Å and requires additional review at the wavelength extremes.
- Order merging uses quality-aware weights, valid overlap only, overlap-only edge tapering and controlled order-level harmonisation. This means more of each order is generally retained and order-merging defects from previous versions are corrected.
- The old production CCF and least-squares CCF workflow has been replaced by regularised least-squares deconvolution (LSD). A delta-function template is no longer used by default.
- Radial velocities are measured from the first moment of the regularised LSD profile, rather than the previous fitted centre of a symmetric rotational profile. Precision has improved significantly.
- The saved `vbroad` is the observed rotation-equivalent line width and includes instrumental broadening. This is equivalent to what was labelled `vsini` in 1.6. The 2.1 `vsini` has the calibrated instrumental broadening removed, but macroturbulence is not removed, so it

should not be interpreted as a pure measurement of projected rotation. This improves `vsini` measurements of narrow-lined stars.

- Post-reduction now uses two conservative cosmic-ray checks: an order-level filter followed by a light-touch check of the merged spectra. Both remove only narrow positive spikes and write audit information. This addresses over-correction by the 1.6 post-processing filter.
- Large datasets (>500 spectra) of a single target can be processed internally in chunks.
- Each target folder now separates reduced frames, processing files, supplementary information and principal scientific products for clarity.
- `final_data` has been simplified. Continuum-fitting and RV/line-width diagnostics are saved separately in `supplementary_data`.

## 3 Installation

### 3.1 Requirements

MEGARA can run on macOS or Linux systems. You will need a licensed version of MATLAB and access to a terminal. All directories from the home directory to the MEGARA files or data files should not have spaces in their names. This version of MEGARA has been tested with the MATLAB release and toolboxes listed in Table 1.

| Software                                | Version                | Core or option |
|-----------------------------------------|------------------------|----------------|
| MATLAB                                  | 24.2 (R2024b) Update 5 | Core           |
| Optimization Toolbox                    | 24.2                   | Core           |
| Signal Processing Toolbox               | 24.2                   | Core           |
| Image Processing Toolbox                | 24.2                   | Core           |
| Statistics and Machine Learning Toolbox | 24.2                   | Core           |
| Curve Fitting Toolbox                   | 24.2                   | Core           |
| Parallel Computing Toolbox              | 24.2                   | Core           |

Table 1: Requirements for MEGARA. Software indicated as “Core” is required for operation.

Full commands are provided in `INSTALLATION_INSTRUCTIONS.txt` in the main MEGARA 2.1 directory. The essential steps are:

1. Install MEGARA 2.1 in its own program directory.
2. Create independent raw-data and reduced-data root directories outside the MEGARA folder.
3. Put `pointer_raw_data_directory.m` in the raw-data root and `pointer_reduced_data_directory.m` in the reduced-data root.
4. Add the complete `Version_2_1` tree and both data roots to the MATLAB path.
5. Compile the bundled `synspec`, `rmod` and `rotin3` programs with `gfortran`. Keep the executables beside `synspec_pointer.m`.
6. Install SUPPNet and its Conda environment, then set the `suppnet_script`, `conda_executable` and environment defaults in `run_suppnet_spectrum.m`.

The HRSP barycentric executable does not need to be installed for MEGARA 2.1.

Organise the raw-data root into subdirectories representing observing periods that can share one flat and wavelength calibration. A directory may represent a night, a run or a short group of runs, but it must contain the appropriate flats and ThAr frames. Periods with a changed fibre, an opened spectrograph tank, a visibly shifted trace or another instrumental change must be separated. As in 1.6, keeping a calibration group no longer than about one month is a useful starting rule, but trace stability is the actual criterion.

Only the raw- and reduced-data pointer files are used in the 2.1 release; the former MEGARA-program pointer is not required. The original 1.6 instruction to change permissions on the HRSP barycentric directory is also obsolete because the correction is now calculated in MATLAB. SynSpec, `rmod` and `rotin3` still need to be compiled as described in the installation text file. On a multicore computer, review MATLAB's Parallel Preferences before a large reduction. Parallel output can appear out of order in the Command Window even though the returned products are ordered.

### 3.2 Recommended directory arrangement

|                              |                         |
|------------------------------|-------------------------|
| MEGARA/Version_2_1/          | program files only      |
| Astronomy_Data/Raw_Data/     | raw HERCULES FITS files |
| Astronomy_Data/Reduced_Data/ | MEGARA outputs          |

Keeping the data outside the release directory prevents observations and reductions from being copied, overwritten or deleted when MEGARA is updated.

## 4 Running MEGARA

Edit and run `User_codes/Run_megara.m`. Set `options.objectname`, select any required observing folders, enter stellar parameters where necessary and review every option. For an existing set of reduced frames, set `options.skip_reduction=true` to rerun post-reduction only.

### 4.1 A standard run

For a normal reduction:

1. Open `User_codes/Run_megara.m`.
2. Set `options.objectname`. An empty target selection processes every stellar target found by the census; the special group names and multi-target structure format are described below.
3. Set `options.reduction_folders` if only particular raw observing folders should be searched. An empty value searches them all.
4. Supply `options.teff`, `options.logg` and the input `options.vsini`, or leave them empty only when reliable database values are available.
5. Review every logical option and its dependencies. In particular, select no more than one new continuum-fitting method.
6. Save and run the script using MATLAB's *Run* button or `run('Run_megara.m')`.

If several targets or frames are processed in parallel, progress messages may be interleaved. The filenames in each message identify the frame being handled; do not interpret the printed order as the final observation order.

### 4.2 Changes to the user interface

| MEGARA 1.6                      | MEGARA 2.1                                       |
|---------------------------------|--------------------------------------------------|
| <code>objectname</code>         | <code>options.objectname</code>                  |
| <code>auto_continuum_fit</code> | <code>options.automatic_continuum_fit</code>     |
| No template mode                | <code>options.template_continuum_fit</code>      |
| <code>cross_correlation</code>  | <code>options.calculate_lsd_profile</code>       |
| <code>rv_measurement</code>     | <code>options.radial_velocity_measurement</code> |
| <code>supress_figures</code>    | <code>options.suppress_figures</code>            |
| <code>overwrite_fulldata</code> | <code>options.overwrite_full_data</code>         |

| MEGARA 1.6     | MEGARA 2.1                                      |
|----------------|-------------------------------------------------|
| Numeric 1 or 0 | Logical <code>true</code> or <code>false</code> |

### 4.3 User inputs and options

All inputs are set in `User_codes/Run_megara.m`. Logical options use `true` for on and `false` for off. The values described as defaults below are the standard starting values; users should still inspect every setting in their copy of the run file before starting a reduction.

#### 4.3.1 Target and stellar inputs

`options.objectname`

Target to process. Use a character vector or string such as `'hd_48501'`. Names are canonicalised to lower case, but users should use a consistent identifier. An empty value processes every stellar target found by the census. The special values `'musician'`, `'tess'` and `'exocomet'` select the corresponding target groups. A structure can be used to supply a named list of targets.

`options.reduction_folders`

Raw-data observing folders to search. Supply a cell array or string array, for example `\{'Jan2018';'Feb2018'\}`. Folder names must match the raw-data subdirectories. An empty value searches all observing folders below the raw-data root.

`options.teff`

Effective temperature in kelvin. Enter one finite scalar or leave empty when the value is available from the MEGARA target database. It is used to select the nearest atmosphere model and to construct synthetic information for continuum fitting and LSD.

`options.logg`

Surface gravity as  $\log_{10}(g)$  in cgs units. Enter one finite scalar or leave empty when the target database supplies it.

`options.v sini`

Input estimate of projected rotational velocity in  $\text{km s}^{-1}$ , used to prepare synthetic spectra and masks. This is an input guide, not the measured MEGARA 2.1 output `vsini`. Enter a scalar or leave empty when a database value is available.

#### 4.3.2 Reduction control

`options.star_census (default true)`

Rebuild the inventory of stellar observations. Use `true` when raw files or observing folders have changed. Once the inventory is current, it may be set to `false` for repeat post-reduction work.

`options.skip_reduction (default false)`

Skip extraction and wavelength calibration of raw FITS frames. Use `true` when valid `J*.mat` products already exist and only post-reduction products need to be rebuilt. MEGARA also skips a target whose identified observations have already been reduced or rejected.

`options.skip_post_reduction (default false)`

Skip continuum application, merging, LSD calculations and derived measurements. This is useful when the immediate aim is only to create reduced frames. Selecting LSD, RV or FAMIAS options at the same time does not override this switch and produces a warning.

`options.skip_manual_file_check (default false)`

Suppress the interactive Good/Bad classification window. Ambiguous frames are retained and reported rather than being rejected interactively. Use with care for unattended reductions and inspect the file-check record afterwards.

`options.blue_data_chop_value (default 500)`

Number of detector pixels removed from the blue end because flat-field coverage is incomplete.

The value must be a finite number greater than or equal to zero. A value of 300 is commonly used when retaining more of the Ca II H and K region, but the affected orders should be inspected.

#### 4.3.3 Dataset size and wavelength range

**options.chunked\_post\_reduction (default false)**

Process already reduced spectra internally in batches to control memory use. This is recommended for a single target with more than roughly 500 spectra. All observations are recombined into the normal final products; this option does not select only part of a dataset. A new manual continuum fit cannot be created during a chunked run.

**options.post\_reduction\_chunk\_size (default 500)**

Positive integer giving the number of reduced spectra processed per chunk when chunking is enabled.

**options.extended\_wavelength\_range (default false)**

Use approximately 3750–8950 Å rather than the standard 3869–6866 Å range. The extreme blue and red regions require additional review. Template continuum fitting is restricted to the standard range.

**options.apply\_barycentric\_correction (default true)**

Apply the MATLAB barycentric correction when building merged final spectra. The wavelength arrays in the reduced J\*.mat frames remain unshifted; their stored correction is used at this stage. If a catalogue target cannot be identified, MEGARA warns the user and uses an available coordinate fallback where possible.

#### 4.3.4 Continuum fitting and order merging

**options.apply\_continuum\_fit (default true)**

Normalise the extracted orders using a saved or newly created continuum fit. This must be **true** when **options.order\_merge** is **true**. If no usable saved fit exists, select exactly one of the three creation methods below.

**options.manual\_continuum\_fit (default false)**

Create a new interactive order-by-order continuum fit. This is the precision route for difficult spectra. It replaces the saved manual fit, so preserve a copy before intentionally rerunning it. It cannot be combined with template or automatic fitting, or created during chunked processing.

**options.template\_continuum\_fit (default false)**

Interpolate the empirical spectral-standard continuum grid using the target temperature, then apply it before order merging. It requires an input or database **options.teff**, **options.order\_merge=true**, and the standard wavelength range. SUPPNet is not run in this mode.

**options.automatic\_continuum\_fit (standard automatic-run value true)**

Run the two-stage automatic method: synthetic-guided order normalisation and quality-aware merging followed by SUPPNet correction of the merged spectrum. It requires **options.order\_merge=true** and cannot be selected with either manual or template fitting.

**options.order\_merge (default true)**

Combine the continuum-normalised echelle orders on the selected common wavelength grid. It requires **options.apply\_continuum\_fit=true**. Regularised LSD, RV measurements and FAMIAS output all require merged spectra.

#### 4.3.5 Rebuilding products and measuring profiles

**options.override\_full\_data (default true)**

Rebuild the continuum-normalised, order-merged spectra in **final\_data**. Set **false** to load an existing valid merged product when only LSD profiles or derived measurements need to be regenerated.

**options.override\_post\_reduction (default true)**

Force post-reduction processing and regeneration of regularised LSD profiles and related measurements. Set **false** to retain valid existing products; MEGARA will still process when a required product is absent or incomplete.

**options.calculate\_lsd\_profile (default true)**

Calculate regularised LSD profiles from the merged spectra, replacing the former CCF products. This requires **options.order\_merge=true**; recalculation overwrites the corresponding regularised-LSD product for the target.

**options.radial\_velocity\_measurement (default true)**

Measure relative and systemic RVs from the first moments of the regularised LSD profiles and calculate **vbroad** and **vsini** from their second moments. This requires **options.calculate\_lsd\_profile=true**. Instrumental broadening is included in **vbroad** and removed from **vsini**; macroturbulence is not removed from either quantity.

**options.extend\_velocities (default false)**

Extend the regularised-LSD velocity grid from  $\pm 200$  to  $\pm 400$   $\text{km s}^{-1}$ . Use this for very broad or strongly shifted profiles; the wider grid increases the calculation size.

#### 4.3.6 Figures and alternative output formats

**options.suppress\_figures (default false)**

Suppress normal diagnostic figures during processing. Interactive manual continuum fitting still requires its fitting display.

**options.ascii\_output (default false)**

Write supported principal results in plain-text form in addition to the MATLAB products. See Section 13 for the output layout.

**options.fits\_output (default false)**

Write supported merged spectra and profile products as FITS files in addition to the MATLAB products.

**options.FAMIAS\_output (default false)**

Write regularised LSD profiles and their times in a FAMIAS-ready format. This requires both **options.calculate\_lsd\_profile=true** and **options.override\_post\_reduction=true**. The output routine rejects non-finite times, profiles or weights rather than writing an invalid file.

The option dependencies are checked before processing begins. MEGARA stops with an explanatory error for incompatible selections.

## 5 Reduction-stage changes

The overall sequence remains target census, file checks, reduction and processing, but several internal operations have changed since 1.6.

- The FITS reader locates the data block from the FITS header instead of relying on a fixed byte offset.
- Spectral traces are sampled at fractional positions, reducing integer-row trace quantisation.
- Flat extraction retains profile-derived weights and a validated extraction-aperture width. Cached flats are rejected when required calibration fields are absent.
- Extracted flat-field spectra are no longer median-smoothed before division into the stellar spectra. The former 30-pixel median filter has been replaced by a one-pixel pass-through, preserving pixel-scale detector and chip-response structure so that it is corrected rather than blurred.
- ThAr and other calibration resources are found using explicit paths instead of assumptions about the current MATLAB directory.
- Header handling, rejected-file handling and restart behaviour are more defensive.

- Multi-frame reduction retains parallel processing but limits worker use to reduce large repeated data transfers. A single frame does not start a parallel pool.
- Completed `J*.mat` products are retained on retry when they pass the completion checks.

These are behavioural changes and a representative target should be compared with its 1.6 reduction before a large archive is migrated.

## 6 Continuum fitting and order merging

### 6.1 Manual continuum fitting

Manual fitting remains the precision route for difficult objects. The fit is made order by order and saved in `supplementary_data`. Existing manual fits should be copied elsewhere before deliberately overwriting them. Invalid continuum values receive zero merge weight.

### 6.2 Automatic continuum fitting

The automatic method is now:

```
synthetic-guided order fit
|
quality-aware order merging
|
SUPPNet merged-spectrum correction
```

The first stage places individual orders on a sufficiently consistent scale for merging. Synthetic information guides the fit while broad stellar lines and unreliable synthetic regions are masked or handled by stable fallbacks. Fits are constrained to remain positive, and endpoint holding or one-sided anchors are used where necessary. SUPPNet then corrects the continuum of the complete merged representative spectrum. SUPPNet is the fully convolutional neural-network continuum-normalisation method introduced by Róžański et al. [8].

The O<sub>2</sub> A-band at 7590–7700 Å is excluded from the coarse normalisation in extended mode and is smoothly bridged for the SUPPNet stage. Continuum products for the standard and extended ranges are cached separately.

### 6.3 Template continuum fitting

Template mode uses an empirical grid of continuum shapes derived from HERCULES observations of 12 spectral standards. The standards span O9V to K5V and were reduced consistently with MEGARA 2.1. Their order-by-order continuum shapes are stored in `spt_continuum_template_grid_v1_standard.mat`. The grid is intended for the standard 3869–6866 Å range only; template mode stops with an error if the extended wavelength option is selected.

| Standard  | Spectral type | Adopted $T_{\text{eff}}$ (K) | Supported interval (K) |
|-----------|---------------|------------------------------|------------------------|
| HD 38666  | O9V           | 33300                        | 26950–39650            |
| HD 52018  | B2V           | 20600                        | 17300–26950            |
| HD 24626  | B7V           | 14000                        | 11650–17300            |
| HD 319    | A1V           | 9300                         | 8605–11650             |
| HD 16555  | A6V           | 7910                         | 7465–8605              |
| HD 109799 | F1V           | 7020                         | 6650–7465              |
| HD 181958 | F7V           | 6280                         | 6105–6650              |
| HD 4307   | G0V           | 5930                         | 5705–6105              |
| HD 10700  | G8V           | 5480                         | 5325–5705              |

| Standard  | Spectral type | Adopted $T_{\text{eff}}$ (K) | Supported interval (K) |
|-----------|---------------|------------------------------|------------------------|
| HD 74576  | K1V           | 5170                         | 5135–5325              |
| HD 191408 | K2V           | 5100                         | 4770–5135              |
| HD 27274  | K5V           | 4440                         | 4110–4770              |

The target must have a finite input `options.teff` between 4110 and 39650 K. If its temperature exactly matches a grid node, MEGARA uses that standard directly. Otherwise, it identifies the nearest grid temperatures below and above the target and calculates the fractional temperature position

$$w = \frac{T_{\text{target}} - T_{\text{lower}}}{T_{\text{upper}} - T_{\text{lower}}}.$$

For every order, the two standard continuum shapes are first interpolated in wavelength onto the target order without extrapolating beyond their valid coverage. MEGARA then forms the linearly interpolated shape

$$C_{\text{template}} = (1 - w)C_{\text{lower}} + wC_{\text{upper}}.$$

Temperatures between the lower supported limit and the coolest grid node use the K5V template; temperatures between the hottest node and the upper supported limit use the O9V template. The temperature is the only quantity used for interpolation: spectral type, surface gravity and projected rotation are descriptive properties of the standards, not additional interpolation axes. Users should therefore inspect results for giants, chemically peculiar stars, emission-line objects or targets whose spectra differ substantially from the main-sequence standards.

The interpolated shape defines relative structure, not the absolute count level of the target. For each order, MEGARA forms the median observed spectrum and scales the template using the 99.5th percentile of the observed-to-template ratio. A template order is accepted only when it has at least 80% valid wavelength coverage, at least 30 valid scale points, a positive finite scale and a stable continuum-shape range. Missing, non-positive or unstable template orders are filled with invalid values and given zero weight in the merge; neighbouring valid orders may still supply coverage.

The accepted continuum is applied directly to each order before the normal quality-aware merge. SUPNet is not run afterwards. Automatic overlap-level harmonisation is also skipped because it would alter the relative order levels defined by the empirical template.

For reproducibility, MEGARA saves the target temperature, grid filename, lower and upper standards, interpolation weight, wavelength coverage, shape range, scale and acceptance status for every order. These are written to `supplementary_data/automatic_continuum_results/` in `<target>_interpolated_spt_template_v1_standard.mat`. The saved fit can be reviewed with `plot_continuum_fits.m`.

## 6.4 Order merging safeguards

MEGARA merges only real extracted coverage. It does not extrapolate beyond an order, and invalid pixels have zero weight. Edge tapering is applied only where a valid neighbouring order overlaps, preserving unique edge coverage. Weights include signal, flat-field quality and extraction quality. Flagged gaps remain gaps instead of being converted into straight-line bridges.

Automatic mode uses small regularised overlap corrections to harmonise relative order levels. Template mode skips this correction because the template defines the relative continuum level. A complete observation is rejected only for a broad coverage or spectral-shape failure; isolated bad pixels do not discard an otherwise useful frame.

## 7 Wavelength range

The standard range is 3869–6866 Å. Setting `options.extended_wavelength_range=true` gives approximately 3750–8950 Å. The extreme blue and red regions can have weaker flat coverage, stronger telluric structure and less secure continuum placement, so extended products require closer inspection. Template continuum fitting is intended for the standard range.

## 8 Barycentric correction

MEGARA 2.1 uses a compile-free MATLAB implementation that is a close translation of the barycentric-correction algorithms in the HERCULES Reduction Software Package (HRSP) written by Jovan Skuljan [2]. The purpose of the translation was to preserve the established HRSP calculation and conventions while removing the need for users to compile and run the original C code. It is thus a MATLAB translation and integration of Jovan Skuljan’s HRSP work.

The implementation uses the bundled DE405 ephemeris together with MATLAB versions of the required stellar-catalogue and Earth-orientation data. Target lookup supports HD, HIP, HR, Bayer and Flamsteed identifiers, including common component mapping. Catalogue positions are propagated using parallax, proper motion and radial velocity. The observatory rotation term includes the HERCULES site position and Earth-orientation model.

The translated calculation was tested against reference values generated by HRSP, including targets with substantial proper motion and radial velocity. This regression comparison is retained with the MEGARA barycentric-correction tests.

If a target is not identified, MEGARA warns and uses coordinates from the FITS header. This fallback does not include catalogue parallax, proper motion or radial velocity and should therefore be checked carefully.

For every reduced frame, `bcorr` is calculated and stored in the `J*.mat` structure. The wavelength values in that file remain in the original extracted frame. During post-reduction, the correction is applied to the wavelength grid only when `options.apply_barycentric_correction=true`. The resulting application state is recorded in the final and optional exported products.

MEGARA 2.1 contains no HRSP executable or C-code dependency. Historical instructions for compiling `barycorr_hrsp` do not apply.

## 9 Post-reduction cosmic-ray filtering

The 1.6 ensemble-median method could mistake genuine absorption or time-variable line structure for cosmic rays. MEGARA 2.1 therefore uses two deliberately conservative post-reduction checks, both of which remove only positive, isolated events.

The first check is applied to the individual continuum-normalised orders before merging. Its baseline is a 31-pixel running median and its noise is estimated locally with a 101-pixel rolling window, rather than assigning one noise value to a whole order. A candidate must exceed  $6\sigma$  locally and, when at least three spectra are available,  $5\sigma$  relative to the other observations. A confirmed event is extended to its positive  $3\sigma$  wings and may contain no more than five pixels. Accepted pixels are replaced by the local median. The audit product `post_reduction_cosmic_ray_cleanup.mat` records every replacement.

A second, light-touch check is applied to the merged spectra before SUPPNet. It requires both an  $8\sigma$  local excess and a  $6\sigma$  excess relative to the other observations, accepts events no wider than three pixels, and is skipped when fewer than three spectra are available. Its replacements are recorded in the `positive_spikes_before_supernet` field of `post_reduction_merged_pixel_cleanup.mat`.

## 10 Regularised LSD profiles

Least-squares deconvolution (LSD) combines the information carried by many photospheric absorption lines into a high signal-to-noise mean profile. It assumes that the selected lines share a common velocity-space shape, scaled by their line-mask weights, and that overlapping line depressions add linearly. The result is therefore a weighted pseudo-line rather than the profile of any one physical transition. These assumptions are most reliable for relatively weak lines and should be treated cautiously for strong lines, severe blends or spectra containing several stellar components [3, 4].

### 10.1 Change from the MEGARA 1.6 calculation

The CCF and least-squares CCF calculations described in the 1.6 manual have been replaced by one regularised LSD inversion. The old route first correlated the absorption spectrum with a weighted delta-function mask, then attempted to remove the mask autocorrelation by a separate deconvolution. Resampling and renormalisation followed, and the adjusted-mask stage could create a second competing profile product. Although this worked well, several sequential operations affected the final baseline, depth and noise behaviour (Figure 1).

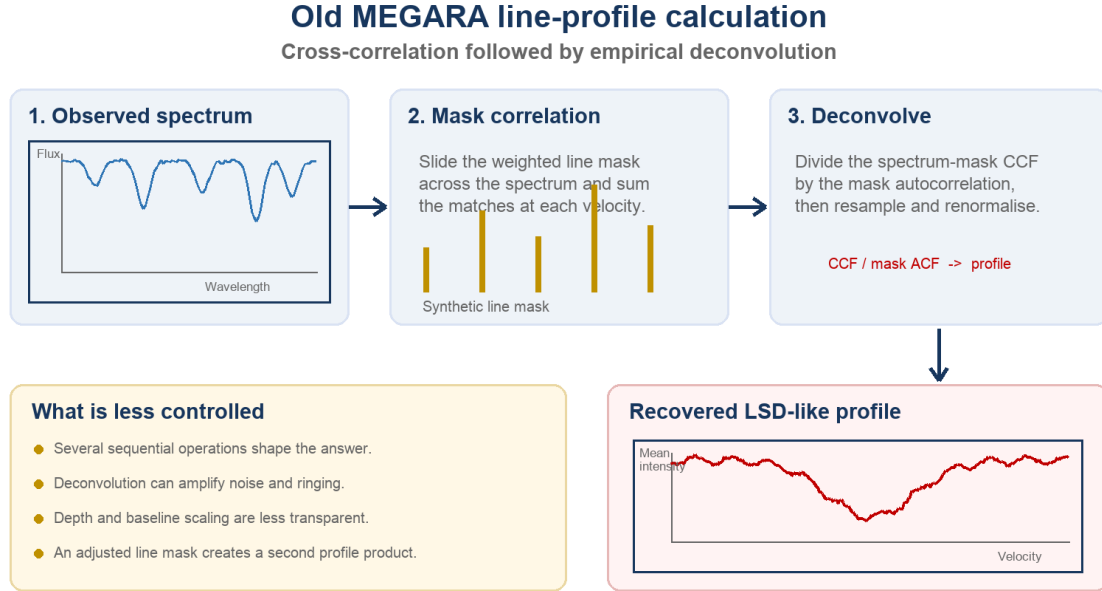


Figure 1: Schematic of the former MEGARA line-profile calculation, reproduced from the MEGARA research notes. Correlation with the line mask was followed by a separate empirical deconvolution and renormalisation.

MEGARA 2.1 instead writes a single forward model for the complete retained spectrum and solves directly for one common mean profile (Figure 2). This is the standard design-matrix view of LSD introduced by Donati et al. and developed in modern regularised and multiprofile implementations [3, 4].

### 10.2 Forward model and design matrix

Let  $y = 1 - I$  be the observed absorption spectrum and let  $z$  be the unknown mean absorption profile sampled on the common velocity grid. MEGARA models the data as

$$y \simeq Az,$$

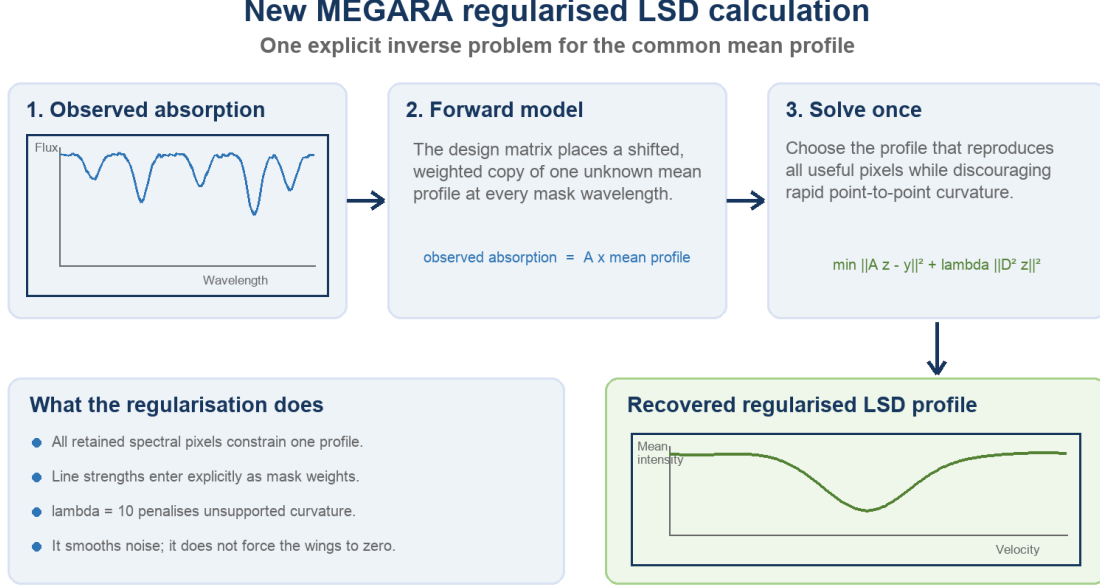


Figure 2: The MEGARA 2.1 regularised-LSD calculation, reproduced from the MEGARA research notes. All retained spectral pixels constrain one common profile in a single regularised inverse problem.

where each row of the sparse design matrix  $A$  represents one retained wavelength pixel and each column represents one LSD velocity bin. For a mask line of rest wavelength  $\lambda_0$  and relative weight  $w$ , the velocity of wavelength pixel  $\lambda_i$  is

$$v_i = c \ln \left( \frac{\lambda_i}{\lambda_0} \right).$$

If  $v_i$  lies between two velocity bins,  $w$  is divided between the two matrix columns by linear interpolation. Thus a pixel at  $+3.2 \text{ km s}^{-1}$  between the  $+2$  and  $+4 \text{ km s}^{-1}$  bins contributes  $0.4w$  and  $0.6w$ , respectively (Figure 3).

The construction is repeated for every retained line. Stronger lines place larger coefficients in  $A$ ; where lines overlap, their coefficients occupy the same row and their predicted absorptions add. Most matrix elements are zero, so MEGARA stores  $A$  as a sparse matrix. Multiplication by a proposed profile then produces the complete model absorption spectrum  $Az$  (Figure 4).

### 10.3 Regularisation and production settings

MEGARA determines  $z$  by minimising

$$\|Az - y\|^2 + \Lambda \|D_2 z\|^2,$$

where the first term requires the profile to reproduce the retained spectral pixels and  $D_2$  is the second-difference operator. The curvature penalty suppresses rapid bin-to-bin structure that is not supported by the data. The production settings are  $\Lambda = 10$ , second-order regularisation and a  $2 \text{ km s}^{-1}$  velocity step. This sampling is not the independent physical resolution: the adopted instrumental FWHM is  $6.993 \text{ km s}^{-1}$ , so one resolution element spans approximately 3.5 LSD bins.

Regularisation is a controlled compromise. It reduces noise amplification and ringing but does not impose a rotational profile, force the wings to zero, separate rotational and macroturbulent

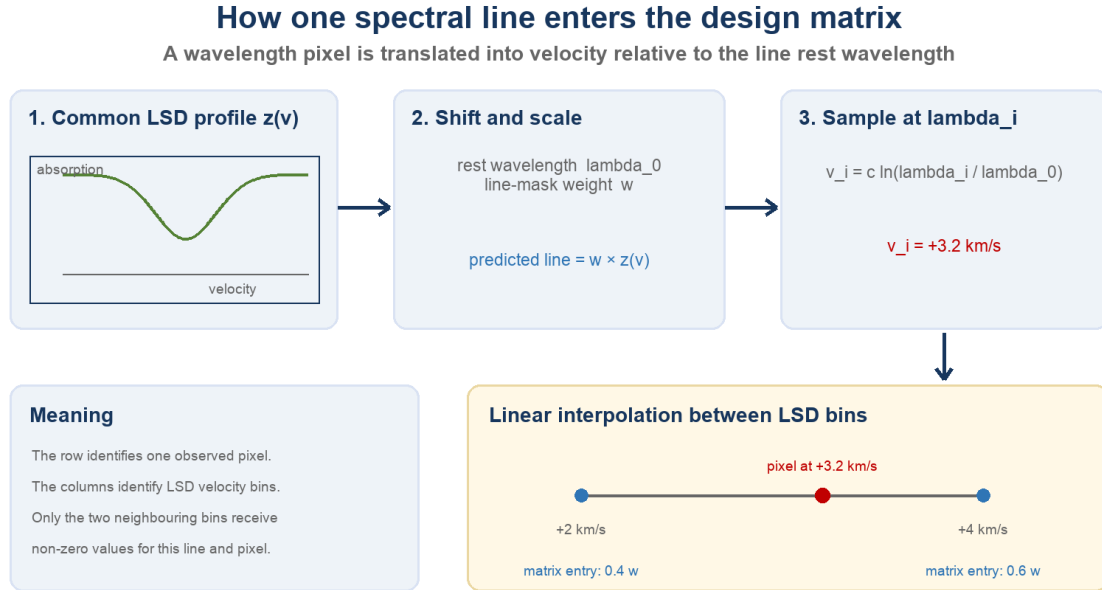


Figure 3: How one mask line contributes to a row of the LSD design matrix.

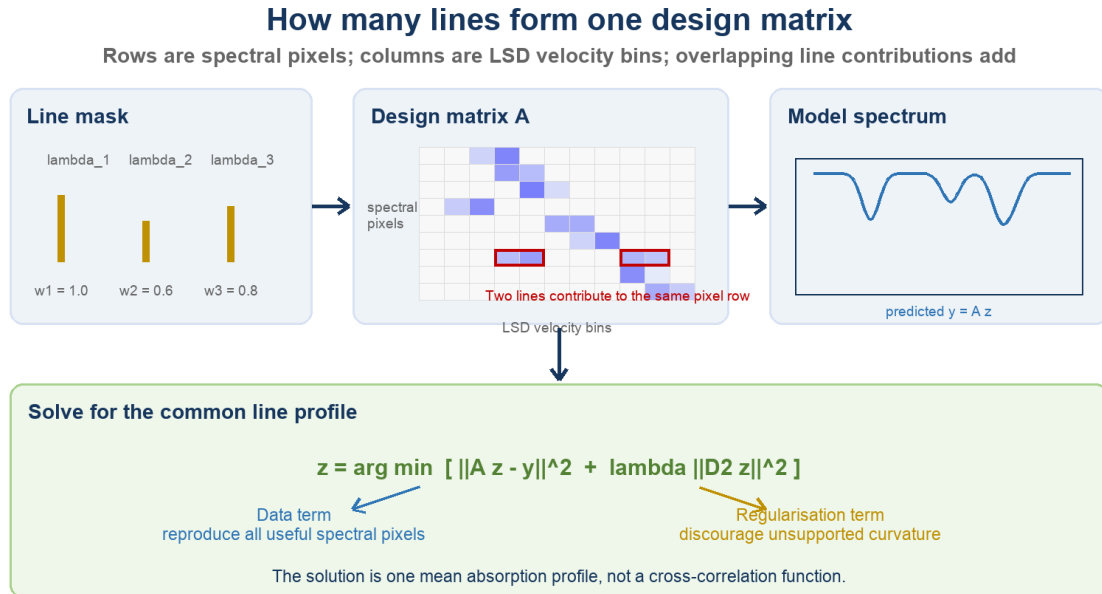


Figure 4: Construction of the sparse design matrix from the full line mask and solution for one common profile.

broadening, or guarantee that every recovered feature is stellar. The production profile is stored using  $I_{\text{LSD}} = 1 - z$ . It is not rescaled to unit depth; only a constant median level measured in the extreme wings is removed. Consequently, profile depths may be compared within a consistently processed series, but they should not be interpreted as the depth of a specific spectral line.

Modern extensions can fit several simultaneous profiles, adjust line strengths and reconstruct denoised spectra. Such methods have been applied successfully to pulsating stars [5]. MEGARA 2.1 uses a simpler single-profile Stokes- $I$  solution with fixed line-mask weights.

The primary product is `regularised_lsd_profile_<target>.mat`. It contains the velocity axis, intensity and absorption profiles, observation times, mask metadata, regularisation settings, velocity convention and baseline information. The extended-velocity option changes the profile range from approximately  $\pm 200$  to  $\pm 400$  km s $^{-1}$ . Optional FAMIAS products are generated from these regularised LSD profiles, not from the old CCF types.

LSD results should be treated with caution for emission-line objects, composite or binary spectra, strong blends, residual telluric or continuum structure, and targets with uncertain stellar parameters. The common-profile assumption can fail when different lines respond differently to abundance, temperature, pressure, magnetic field or pulsation. In particular, an LSD profile should not automatically be treated as a physically real spectral line [4].

A future update of MEGARA will have a daughter code with more options for generating and testing LSD profiles using different methods.

## 11 Radial velocity and line-width measurements

### 11.1 First-moment radial velocity

MEGARA 2.1 no longer adopts the fitted centre of a symmetric rotational profile as its production RV. It fits a local linear baseline to the LSD profile and calculates the first moment of the measured absorption. The integration half-width is selected from the median profile using a profile-containment rule, then fixed for every observation of that star. Each observation is iteratively recentered until the moment converges.

This use of velocity moments follows the moment-method framework for stellar line-profile variability described by Aerts, De Pauw and Waelkens [7]. MEGARA uses the first moment here as an RV estimator; it does not perform pulsation-mode identification.

The systemic velocity is the first moment of the median profile. The saved relative RV for an observation is its first moment minus the median-profile moment. After this measurement, MEGARA subtracts the systemic velocity from the saved LSD velocity axis so that the median profile is centred at zero in the stellar rest frame. The removed absolute value remains available in `systemic_velocity_direct`. Formal errors propagate noise measured in the LSD wings and include a conservative fraction-of-sampling floor.

### 11.2 Meaning of the saved `vsini` and `vbroad` values

The second moment measures the observed width of the LSD profile. MEGARA converts the observed second moment to a rotation-equivalent width and saves it as `vbroad`. This quantity includes instrumental broadening.

For `vsini`, MEGARA subtracts the calibrated instrumental variance before converting the remaining second moment to a rotation-equivalent width. Thus `vbroad` includes the instrumental contribution whereas `vsini` does not. For the adopted rotation-variance coefficient of 0.225, the two saved quantities are related by

$$(v \sin i)^2 = \max\left(0, v_{\text{broad}}^2 - \frac{\sigma_{\text{inst}}^2}{0.225}\right).$$

The release calibration has an instrumental FWHM of  $6.993 \text{ km s}^{-1}$ , corresponding to a Gaussian sigma of  $2.970 \text{ km s}^{-1}$  and a resolving power of approximately 42,869. The calibration was measured from Gaussian fits to 106,990 accepted extracted ThAr lines in 88 frames.

Macroturbulence is not modelled or removed from either quantity. Consequently `vsini` should not be treated as a pure projected rotational velocity, particularly for stars where macroturbulence is significant. Pulsation, binarity, spots, line asymmetry and other line-shape contributions can also affect both measurements. The detailed moments, calibration information and interpretation are stored in the separate diagnostics file.

## 12 Measured performance and development validation

The following values summarise representative measurements made while developing and testing MEGARA 2.1. They describe the tested CCD486 data and the present reduction settings; they are not guaranteed performance limits for every target or observing run. In particular, empirical scatter can include stellar variability, blends, continuum errors, calibration changes and imperfect literature reference velocities.

The HERCULES instrument and its available resolving-power configurations are described by Hearnshaw et al. [6]. The measurements below refer specifically to the CCD486 configuration and MEGARA 2.1 test sample.

| Quantity                                    | Validation sample                                                              | Result and interpretation                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instrumental broadening and resolving power | 106,990 accepted extracted ThAr lines from 88 CCD486 frames                    | Median FWHM $6.993 \text{ km s}^{-1}$ , Gaussian $\sigma = 2.970 \text{ km s}^{-1}$ and equivalent $R \simeq 42,869$ . This global sigma is the value removed in quadrature from the second moment when calculating $v \sin i$ .                                                                                                                                                                        |
| Wavelength dependence of resolution         | 250-Å bins within the standard 3869–6866 Å range                               | Median binned FWHM spans approximately $6.89\text{--}7.19 \text{ km s}^{-1}$ , equivalent to $R \simeq 41,700\text{--}43,500$ . The global correction is therefore representative rather than a complete wavelength-dependent line-spread model.                                                                                                                                                        |
| Regularised-LSD sampling                    | Production profile grid                                                        | The $2 \text{ km s}^{-1}$ grid gives about 3.5 samples across the measured instrumental FWHM. Adjacent bins are consequently not independent resolution elements.                                                                                                                                                                                                                                       |
| Barycentric calculation                     | Six targets spanning demanding proper-motion and radial-velocity cases         | The largest numerical difference between the MATLAB translation and precision HRSP output was approximately $6 \times 10^{-6} \text{ m s}^{-1}$ in the tested cases. This demonstrates agreement with the translated HRSP algorithm; it is not an absolute accuracy estimate for the ephemeris, catalogue data or observation time.                                                                     |
| Formal RV uncertainty                       | 338 finite epochs in the currently saved 2.1 final files for five RV standards | The median formal error is $100 \text{ m s}^{-1}$ and the observed range is $100\text{--}172 \text{ m s}^{-1}$ . Most measurements reach the deliberately conservative floor of 5% of one $2 \text{ km s}^{-1}$ LSD bin.                                                                                                                                                                                |
| Empirical RV repeatability                  | The same five standards                                                        | Robust production scatter is $18.9 \text{ m s}^{-1}$ for $\beta$ Cet, $51.3 \text{ m s}^{-1}$ for $\beta$ Lep and $23.4 \text{ m s}^{-1}$ for $\beta$ Crv. The much larger values of $292.3$ and $162.7 \text{ m s}^{-1}$ for $\alpha$ TrA and $\beta$ Aqr show that the time-series scatter is strongly target dependent. A focused 23-spectrum $\beta$ Cet run reached $12.8 \text{ m s}^{-1}$ .      |
| Formal $v \sin i$ uncertainty               | The same 338 standard-star epochs                                              | The median propagated uncertainty is $0.132 \text{ km s}^{-1}$ ; the central 68% lies at approximately $0.118\text{--}0.154 \text{ km s}^{-1}$ . Per-star medians span $0.062\text{--}0.214 \text{ km s}^{-1}$ and the complete range is $0.054\text{--}0.449 \text{ km s}^{-1}$ . These formal values do not include uncertainty from macroturbulence or the physical interpretation of the LSD width. |

## Additional pipeline tests

**Order-overlap consistency** In 244 HD 109799 spectra, order-level harmonisation reduced the median absolute overlap mismatch from 0.32% to 0.08%, and the 95th percentile from 1.93% to 0.41%.

**Post-reduction cosmic-ray discrimination** In archived HD 79469 data and synthetic injection tests, candidate changes fell from 15,667 with the old ensemble filter to 750 with the conservative 2.1 filter; in order 91 the count fell from 1,275 to 5. Synthetic tests retained real absorption, variable absorption and narrow emission while removing all injected cosmic rays. A later comparison of 312 matched HD 135825 observations found a median of 9 narrow positive candidates per spectrum in the 1.6 final data and 3 in 2.1; the matched totals were 3,180 and 1,150 respectively. The light-touch merged pass identified only 25 additional pixels in 13 events across 8 of the 324 spectra. The saved audit files should still be inspected for unusual targets.

Formal uncertainty and empirical repeatability answer different questions. The formal RV error describes propagated profile noise subject to the sampling floor; it does not predict the scatter of a variable or poorly modelled star. Conversely, scatter below the formal floor in a stable time series demonstrates repeatability but does not make the reported per-epoch error smaller. The recorded five-standard comparison also found absolute median offsets from literature values ranging from about 5 to 513 ms<sup>-1</sup>. Absolute accuracy should therefore be assessed with appropriate standards from the same instrumental configuration rather than inferred from internal repeatability alone.

## 13 Output directories and files

Each target is stored under the reduced-data root. A typical directory is:

```
Reduced_Data/<target>/
|-- final_data_<target>.mat
|-- regularised_lsd_profile_<target>.mat
|-- reduced frames/
|   |-- J*.mat
|   `-- processing_files/J*_prc.mat
`-- supplementary_data/
    |-- diagnostics_<target>.mat
    |-- star_database_<target>.mat
    |-- automatic_continuum_results/
    |-- post_reduction_cosmic_ray_cleanup.mat
    `-- other audit and cache products
```

### 13.1 Summary of output files

| File or directory                         | Created when | Contents and use                                                                                                                                           |
|-------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reducedframes/J*.mat                      | Reduction    | Extracted, wavelength-calibrated orders and observation metadata. Contains <code>bcorr</code> , but its wavelength arrays are not barycentrically shifted. |
| reducedframes/processing_files/J*_prc.mat | Reduction    | Extraction settings, intermediate measurements and per-frame processing information.                                                                       |

| File or directory                                                       | Created when                            | Contents and use                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>final_data_&lt;target&gt;.mat</code>                              | Merged post-reduction                   | Principal merged spectral product, including wavelength, intensity, times and concise science measurements selected for ordinary use. When RV measurement is enabled it contains <code>vbroad</code> , including instrumental broadening, and instrument-corrected <code>vsini</code> , with their uncertainties. |
| <code>final_data_extended_&lt;target&gt;.mat</code>                     | Extended mode                           | Principal merged product for the extended wavelength range.                                                                                                                                                                                                                                                       |
| <code>regularised_lsd_profile_&lt;target&gt;.mat</code>                 | LSD enabled                             | Velocity axis, regularised LSD profiles, observation times and LSD metadata.                                                                                                                                                                                                                                      |
| <code>supplementary_data/diagnostics_&lt;target&gt;.mat</code>          | Post-reduction                          | Detailed continuum-fitting and RV/line-width diagnostics removed from the simplified <code>final_data</code> structure.                                                                                                                                                                                           |
| <code>supplementary_data/star_database_&lt;target&gt;.mat</code>        | Census/reduction                        | Observation filenames, dates, times, exposure information and source folders.                                                                                                                                                                                                                                     |
| <code>supplementary_data/automatic_continuum_results/</code>            | Automatic or template fitting           | Cached continuum fits, SUPPNet inputs and outputs, and continuum audit products.                                                                                                                                                                                                                                  |
| <code>supplementary_data/post_reduction_cosmic_ray_cleanup.mat</code>   | New merged spectrum built               | Audit of pixels considered and replaced by the conservative order-level post-reduction cosmic-ray filter.                                                                                                                                                                                                         |
| <code>supplementary_data/post_reduction_merged_pixel_cleanup.mat</code> | New merged spectrum built               | Audit of non-physical merged pixels and the light-touch positive-spike check applied before SUPPNet.                                                                                                                                                                                                              |
| <code>supplementary_data/post_reduction_order_edge_coverage.mat</code>  | New merged spectrum built               | Coverage and order-edge information used during post-reduction.                                                                                                                                                                                                                                                   |
| <code>supplementary_data/chunked_post_reduction_files/</code>           | Chunking enabled                        | Temporary or cached chunk products used for large datasets.                                                                                                                                                                                                                                                       |
| ASCII products                                                          | <code>options.ascii_output=true</code>  | Text versions of spectra and summary information for non-MATLAB use.                                                                                                                                                                                                                                              |
| FITS products                                                           | <code>options.fits_output=true</code>   | FITS versions of merged spectra and selected metadata.                                                                                                                                                                                                                                                            |
| FAMIAS products                                                         | <code>options.FAMIAS_output=true</code> | Individual regularised LSD profiles and their times file formatted for FAMIAS.                                                                                                                                                                                                                                    |

## 13.2 MATLAB workspace outputs

As MEGARA runs, the Command Window reports the current target, census and reduction status, atmosphere-model selection, post-reduction stages and any warnings. On return, `Run_megara.m` provides three workspace outputs:

`final_data`

A structure containing the final result for each requested target. For a multi-target run, target names form the outer fields. The contents depend on the selected options. A normal merged

result includes `fullwave`, `fullint`, `jd` and `signal_to_noise`. If LSD and RV/line-width measurements are enabled it also contains the regularised LSD product, relative RVs and errors, systemic velocities, `vbroad`, `vsini` and their errors. Detailed fit and moment diagnostics are moved to the separate diagnostics file described below.

**options.objectname**

The canonical target selection returned by MEGARA. A single target is normally a character vector; a named multi-target request is retained as a structure.

**star\_database**

The census structure used for the run. It records the target association, source filename, exposure time, observation date, Julian date and observing folder where those values are available. The saved version should be used to trace each processed epoch back to its raw file.

The workspace variables are convenient for immediate inspection, but the saved files are the persistent scientific record. A successful run should not be documented only by saving the MATLAB workspace.

### 13.3 Raw-folder inventories, rejected files and calibrations

Several outputs belong to an observing-run directory rather than to one stellar target:

- `file_information.txt` is a human-readable inventory of the FITS files and useful headers in a raw observing folder. `file_info.mat` stores the corresponding MATLAB structure used by the census and file-check routines.
- `badfiles_general.mat` records files rejected by general checks, including files that are incomplete or outside the allowed size range. `badfiles_calibration.mat` records rejected flat and ThAr frames, and `badfiles_<target>.mat` records rejected science frames. Rejected raw files are moved into that observing folder's `removed_files/` directory, rather than deleted.
- `flat_info_blue_<X>.mat` is the cached master-flat result for that observing folder, where `<X>` is the selected blue-pixel chop. MEGARA 2.1 keeps the unsmoothed extracted flat spectrum used for division.
- `ThAr_info_blue_<X>.mat` stores the ThAr wavelength-calibration result for the same blue-pixel chop. A cache is reused only when its required fields and settings pass the current validation checks.

The reduced-data root also contains `ObservationSummaries/`, which holds census products used across targets. Target-specific copies and later diagnostics are collated under the target's `supplementary_data/` directory.

### 13.4 Target MATLAB files in detail

For every accepted raw stellar `J*.fit` frame, MEGARA writes:

- `reducedframes/J*.mat`: extracted and wavelength-calibrated orders, with per-frame metadata such as JD, barycentric correction and signal-to-noise. The barycentric correction is stored but the order wavelength arrays in this file are not shifted.
- `reducedframes/processing_files/J*_prc.mat`: the reduction settings and intermediate information needed to audit how that frame was processed.

The principal merged product is `final_data_<target>.mat`, or `final_data_extended_<target>.mat` for the extended wavelength range. Its core fields are:

**fullwave**

Common wavelength axis in Å. It is barycentrically corrected only when `options.apply_barycentric_correction=true` was used to build the product.

**fullint**

Continuum-normalised, order-merged intensities, with one row per observation in the same order as `jd`.

**jd and signal\_to\_noise**

Observation times and the corresponding per-frame signal-to-noise estimates.

**radial\_velocity and radial\_velocity\_error**

Relative first-moment RVs and their per-epoch errors in  $\text{km s}^{-1}$ . **systemic\_velocity\_direct** records the first moment of the median LSD profile; **systemic\_velocity\_gauss** is retained as a Gaussian-centre diagnostic.

**vbroad, vbroad\_error, vsini and vsini\_error**

Per-epoch second-moment line-width measurements in  $\text{km s}^{-1}$ . The **vbroad** value includes instrumental broadening. The 2.1 **vsini** has the calibrated instrumental variance removed, but macroturbulence is not removed from either quantity. Median, error and scatter summaries are stored in the corresponding **single\_\*** fields.

**regularised\_lsd\_profile\_<target>.mat** stores the stellar-rest-frame LSD velocity axis, profile intensities, JDs and calculation metadata. The **systemic\_velocity\_removed** metadata field records the velocity subtracted from that axis. It replaces the 1.6 **ccf\_<target>.mat** product; the two are not scientifically interchangeable. Continuum internals, baseline choices, moment windows, instrumental calibration and quality information are saved in **supplementary\_data/diagnostics\_<target>.mat**. Census, continuum-cache, cosmic-ray and order-edge audit files are also kept in **supplementary\_data/** as listed in the summary table.

### 13.5 Signal-to-noise estimate

The signal-to-noise value saved in each reduced frame and propagated into **final\_data** is measured from extracted order 100, at approximately 5777 Å. MEGARA converts the extracted signal from ADU using the calibration in **photon\_ADU\_conversion.mat**. The exposure-dependent dark estimate and the stored readout/bias calibration are included in the background term. For source signal  $L$  and background contribution  $B$ , the sampled signal-to-noise spectrum is

$$S/N = \frac{L}{\sqrt{L + 2B}}.$$

The saved scalar is the rounded median over pixels 1700–2219 of the available order-100 extraction; if the order is shorter, the full available region is used. A missing or non-finite result is stored as NaN. This is a representative per-frame value, not a wavelength-dependent error spectrum.

### 13.6 Figures produced during a run

When **options.suppress\_figures=false**, the selected processing stages can produce the following diagnostic figures:

- a full order-merged spectrum plot for every processed target, or a sample order-87 plot when an unmerged product is being inspected;
- all regularised LSD profiles against velocity when **options.calculate\_lsd\_profile=true**;
- relative RV with errors against JD when **options.radial\_velocity\_measurement=true**; and
- instrument-corrected **vsini** with errors against JD when finite measurements are available.

Manual continuum fitting always uses its interactive fitting display. Figures are diagnostic views and are not a substitute for the saved arrays. The legacy tutorial plots are reproduced in the tutorial appendix with explicit notes about changes to the 2.1 calculations.

### 13.7 ASCII output

With **options.ascii\_output=true**, MEGARA writes one **J\*\_reduced.dat** file per observation in the target directory. A merged file has two columns: wavelength in Å and intensity. An

unmerged file has four columns: wavelength, intensity, relative weight and order number.

The accompanying `summary_file_<target>.txt` contains, where available: original filename, JD, observation date, exposure start, exposure duration, flux-weighted midtime, barycentric correction, a flag recording whether the correction was applied, and signal-to-noise. When RV measurement is enabled it also contains systemic velocity, RV, RV error, `vsini`, `vsini_error`, `vbroad` and `vbroad_error`. Velocities are in  $\text{km s}^{-1}$ .

When LSD calculation is enabled, `regularised_lsd_profile_<target>.dat` is also written. Its first column is velocity in  $\text{km s}^{-1}$  and each subsequent column is one LSD profile in observation order.

### 13.8 FITS output

With `options.fits_output=true`, MEGARA writes one `J*_reduced.fits` image per observation. A merged image has two columns, wavelength in Å and intensity. An unmerged image has four columns: wavelength, intensity, relative weight and order number.

Valid metadata are written as FITS keywords; missing non-finite optional values are omitted rather than passed to CFITSIO. Common keys are:

| Keyword         | Meaning                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------|
| FILENAME        | Original frame filename or root.                                                                      |
| JD              | Reduced mid-observation Julian date.                                                                  |
| DATE            | Observation date.                                                                                     |
| START           | UTC exposure-start time.                                                                              |
| EXPTIME         | Exposure duration in seconds.                                                                         |
| FWMT            | UTC flux-weighted exposure midtime.                                                                   |
| BCORR           | Stored barycentric-correction velocity.                                                               |
| BCAPP           | Flag recording whether the correction was applied.                                                    |
| SNR             | Representative signal-to-noise value.                                                                 |
| SYSVEL          | Systemic velocity in $\text{km s}^{-1}$ .                                                             |
| RV, RVERR       | Relative RV and error in $\text{km s}^{-1}$ .                                                         |
| VBROAD, VBRDERR | Observed line width, including instrumental broadening, and its error in $\text{km s}^{-1}$ .         |
| VSINI, VSIERR   | Instrument-corrected line width, with macroturbulence retained, and its error in $\text{km s}^{-1}$ . |

The combined LSD image is written as `regularised_lsd_profile_<target>.fits`. Its first column is velocity and the remaining columns are the observation profiles. Its header identifies the product as regularised inverse-problem LSD and, when available, records the normalisation, velocity step, regularisation strength and order, and number of mask lines.

### 13.9 FAMIAS output

With `options.FAMIAS_output=true`, MEGARA creates `Regularised_LSD_FAMIAS_files/` inside the target directory. It writes one profile file per observation plus `times_<target>.txt`, which lists the files for import into FAMIAS. Weights are estimated from the noise in the first part of each merged spectrum. The writer requires finite times, velocity coordinates, profiles and weights and stops rather than writing malformed FAMIAS files. This output contains only the regularised LSD profiles; the two legacy 1.6 CCF directories are no longer produced.

The exact set of fields depends on the selected options. The principal `final_data` structure is intentionally smaller than in 1.6. Users who need the fit method, tested moment widths, baselines, formal errors, quality flags or continuum audit information should load `supplementary_data/diagnostics_<target>.mat`.

## 14 Migrating reductions from MEGARA 1.6

1. Install 2.1 separately and retain the original 1.6 installation.
2. Keep all raw and reduced data outside both program directories.
3. Configure the new raw and reduced pointer files.
4. Compile the bundled SynSpec executables and install SUPPNet.
5. Translate the old run settings into the `options` structure.
6. Test a small representative target before processing a large archive.
7. Let MEGARA create or migrate the 2.1 target directory layout.
8. Rebuild `final_data` to obtain the new continuum, merge and cosmic-ray treatment.
9. Recalculate regularised LSD profiles and RV/line-width measurements.
10. Do not treat 1.6 CCF products as equivalent to 2.1 regularised LSD profiles.
11. Retain the old products and record the MEGARA version used for every scientific dataset.

## 15 Other useful code

The release contains a deliberately small set of supplementary utilities:

`backfill_bcorr.m` Adds missing MATLAB barycentric corrections to reduced frames without changing existing finite corrections.

`batch_move_files.m` Example metadata-based raw-file mover. Review its selection before use because it changes file locations.

`calculate_resolving_power.m` Estimates resolving power from a suitable CCD486 ThAr reduction. A minimal call is `[summary,lines,orders]=calculate_resolving_power('J9236023.fit')`. By default the utility looks beside the ThAr image for the matching `flat_info` and `ThAr_info` caches. It returns the summary, accepted line measurements and order measurements and produces diagnostic plots. It is not called during a normal reduction.

`display_directory_file_information.m` Displays useful FITS header and file information for a raw-data directory and writes `file_information.txt`. Run it from the directory to be inspected. Its columns include filename, HERCULES exposure type, stored mid-observation time, fibre, object, signal-to-noise where available, and exposure time.

`extract_fullwave_fullint.m` Extracts merged wavelength and intensity arrays from saved final products.

`fix_wonky_shoulders.m` Interactive maintenance tool retained for legacy CCF products; it is not part of the regularised-LSD workflow.

`hercules_fitsread.m` Reads HERCULES FITS images and headers. For example, `[image,header]=hercules_fitsread('J7731001.fit')` returns the oriented detector image and header representation. This helper is retained for compatibility; production reduction uses the current validated FITS-reading path.

`make_new_thardat.m` Interactive maintenance tool for rebuilding ThAr reference positions after an instrumental change such as opening the tank. Edit its input ThAr frame and proposed output name, run it beside `thar.dat`, inspect the overlay and corrections, and review the new MAT file before installing it. The date/configuration lookup must also be updated so that reductions select the new reference only for the intended data.

`make_synspec_file.m` Exports a SynSpec spectrum and line list as text.

`plot_continuum_fits.m` Reviews saved manual, automatic or template continuum fits using the 2.1 target layout.

`plot_fits_image.m` Displays a FITS image with useful scaling. It accepts either a HERCULES FITS filename or an already loaded numeric image, for example `plot_fits_image('J7733017.fit')`.

The 1.6 dark, photon/ADU, manual-CCF and smoothed-CCF utilities are not distributed as

ordinary 2.1 user tools. Their calibration roles are handled by the retained pipeline resources, and the former CCF analysis has been replaced by regularised LSD and moment measurements. Do not copy those legacy utilities into the 2.1 program path merely to reproduce an old workflow.

## 16 Troubleshooting

**Path or cd errors** Confirm that the program and both data roots are on the MATLAB path and that the pointer files resolve to the intended directories. Avoid spaces in the directory chain. If a folder has been renamed or moved, remove the stale MATLAB path entry and add the new one.

**Pointer not found** Confirm that the correct pointer file is in each data root and that both roots are on the MATLAB path.

**SynSpec not found** Check `whichsynspec_pointer`, confirm that the executables are beside it and confirm that the directory is writable.

**SUPPNet fails** Check the Conda executable, environment name and full path to `suppnet.py` in `run_suppnet_spectrum.m`. Test the same command from a terminal.

**Unknown barycentric target** Check the target spelling and FITS-header coordinates. The warning means the calculation lacks catalogue parallax, proper motion and radial velocity.

**Too few ThAr lines** Inspect the flat and ThAr traces for a shift. Separate data taken with different instrumental configurations into different observing-run directories. The 1.6 tutorial typically identified more than 900 lines and retained more than 800 per ThAr frame; these are historical guidance values rather than universal acceptance thresholds. Compare all frames within the same run and investigate any clear outlier.

**Continuum mismatch** Inspect the saved fit with `plot_continuum_fits.m`. Consider manual or template mode if the automatic route is unsuitable for the target.

**Uneven LSD shoulders** First inspect the merged spectrum, continuum audit and line mask. A mismatch can arise from continuum placement, binary components, emission, abundance differences or an unsuitable atmosphere model. The legacy `fix_wonky_shoulders.m` changes old CCF products and is not a repair for regularised LSD profiles.

**Unexpected cosmic-ray changes** Inspect `post_reduction_cosmic_ray_cleanup.mat` for the order-level pass and `post_reduction_merged_pixel_cleanup.mat` for the light-touch merged pass. Both filters should flag only narrow positive spikes.

**Large dataset stalls** Enable chunked post-reduction and reduce the chunk size. Completed reduced frames will normally be retained on retry.

**Interrupted or repeated run** Check which `J*.mat`, merged and LSD products already exist before changing overwrite flags. MEGARA retains completed reduced frames that pass its checks. To repeat post-reduction only, use `options.skip_reduction=true` and leave `options.skip_post_reduction=false`.

**ASCII, FITS or FAMIAS output fails** Confirm that the relevant MATLAB product exists and contains finite times, profiles and metadata. Missing optional FITS keyword values are omitted, but non-finite FAMIAS profiles or weights are rejected. Rebuild the MATLAB product before retrying the export.

## 17 Notes on times

HERCULES headers contain several related time values. MEGARA records the observation date, exposure start, exposure duration and flux-weighted mid-exposure time where available. The barycentric correction uses the appropriate flux-weighted mid-exposure time and time-scale conversion.

Historically, the HERCULES `MJD-OBS` keyword is described in the reduction code as a corrected, heliocentric, mid-observation Julian date. It is not necessarily a conventional Modified

Julian Date formed by subtracting 2,400,000.5. MEGARA stores this header value as `jd`; if it is absent, the header-check routine attempts to reconstruct the mid-observation value from `DATE-OBS`, `REC-STRT` and the flux-weighted offset `HERCFWMT`. The latter is interpreted in seconds.

The barycentric calculation is a separate operation. MEGARA constructs the UTC flux-weighted mid-exposure time, converts it to the time scale required by the translated HRSP calculation, and stores that correction in the reduced frame. The correction is applied to wavelengths only when the final post-reduction spectrum is built. Thus `jd`, `midtime_utc`, `midtime_tt` and `bcorr` describe related but distinct quantities.

For precision timing work, inspect the time fields in the reduced frame and the exported headers, and retain the original FITS header for provenance.

## A Tutorial: reducing the HD 48501 example data

This tutorial updates the worked example supplied with MEGARA 1.6. The example contains 15 observations of HD 48501 obtained in December 2006. A copy of the `tutorial_data` directory is included in the MEGARA 2.1 release. Before running the tutorial, copy that complete directory to your raw-data root. Run the copied dataset from the raw-data root; do not use or generate reduction products inside the MEGARA program directory.

The exact order of messages printed by MATLAB can vary when parallel workers are used. Line counts and fitted values can also change slightly as the reduction methods are updated.

### A.1 Preparation

Complete the installation and pointer-file setup described in this manual. The raw-data root should contain the tutorial observing-run directory and `pointer_raw_data_directory.m`; the independent reduced-data root should contain `pointer_reduced_data_directory.m`. Keep both roots on the MATLAB path.

Open `User_codes/Run_megara.m` and set the following values. Giving the observing-run name explicitly prevents other HD 48501 observations in the raw-data archive from being included.

```
options.objectname = 'hd_48501';
options.reduction_folders = {'tutorial_data'};
options.teff = [];
options.logg = [];
options.vsini = [];
```

```
options.star_census = true;
options.skip_reduction = false;
options.skip_post_reduction = false;
options.skip_manual_file_check = false;
```

```
options.chunked_post_reduction = false;
options.extended_wavelength_range = false;
options.blue_data_chop_value = 300;
options.apply_barycentric_correction = true;
```

```
options.apply_continuum_fit = true;
options.manual_continuum_fit = false;
options.template_continuum_fit = false;
options.automatic_continuum_fit = true;
options.order_merge = true;
```

```
options.override_full_data = true;
options.override_post_reduction = true;
options.calculate_lsd_profile = true;
options.radial_velocity_measurement = true;
options.extend_velocities = false;
options.suppress_figures = false;
```

```
options.ascii_output = false;
options.fits_output = false;
options.FAMIAS_output = false;
```

The automatic continuum option requires a working SUPPNet installation. If SUPPNet is not installed, use a previously saved manual continuum fit or select the template-continuum option after supplying a suitable effective temperature. For this HD 48501 example, suggested template-continuum inputs are `options.teff=7000 K` and `options.logg=4.0`, with the continuum options changed as follows:

```
options.teff = 7000;
options.logg = 4.0;
options.template_continuum_fit = true;
options.automatic_continuum_fit = false;
```

The temperature places the target close to the 7020 K F1V standard in the empirical template grid. These are suggested working values for the tutorial, not new determinations of the stellar parameters. Do not enable more than one method for making a new continuum fit. The empty stellar parameters in the default automatic example are retained from the original tutorial; MEGARA uses its target information and nearest available atmosphere model where possible. Instructions for creating a new manual fit are given in Section A.6.

Run the script with the MATLAB *Run* button or by entering `run('Run_megara.m')` while the script is open.

## A.2 Census and file classification

MEGARA searches the selected observing run for HD 48501 and compares the result with the target's reduced-data directory. For an unused reduced-data root, the census should find 15 stellar observations and no previously reduced observations. Existing products are recognised and are not reduced again unless the applicable overwrite option requests later processing.

With `options.skip_manual_file_check=false`, MEGARA asks the user to review files whose classification is uncertain. Compare the vertical trace of the queried image with the examples of a flat field, ThAr exposure and stellar exposure (Figure 5). Accept a frame only when its proposed classification and quality are credible.

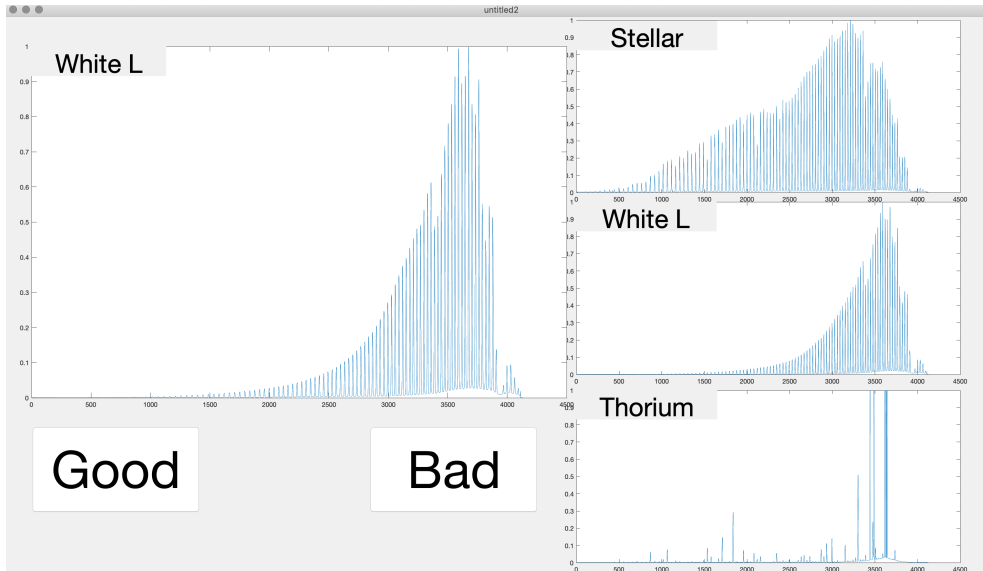


Figure 5: The tutorial file-classification window. The large panel shows the queried frame and the smaller panels provide representative flat-field, ThAr and stellar traces.

MEGARA then presents a summary of the retained and rejected traces (Figure 6). Inspect it before allowing the reduction to continue. The tutorial includes two incomplete files; the file

checks should reject these rather than attempting to reduce them. Rejected raw files are placed in the run's `removed_files` directory and remain available for inspection.

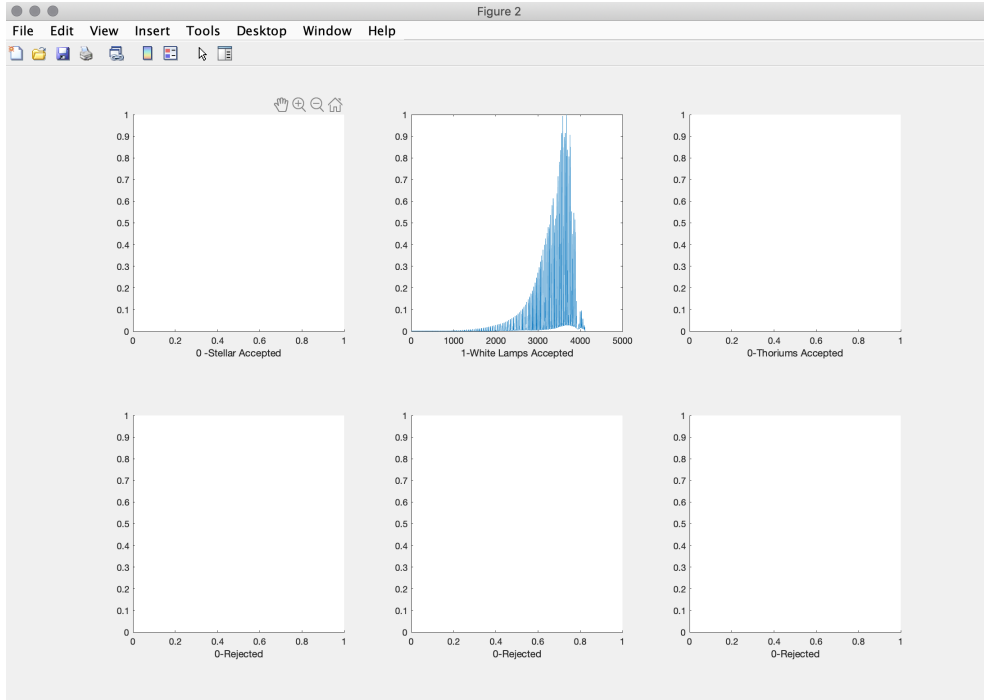


Figure 6: The tutorial file-check summary. The columns correspond to stellar, flat-field and ThAr frames; retained and rejected traces are shown separately.

If a trace is misclassified, the group appears displaced, or an unexpected set of files is rejected, stop the run and investigate the raw directory.

### A.3 Flat-field and ThAr calibration

MEGARA combines the accepted flat exposures, traces the echelle orders, fits the background and extracts the master flat. The calibration information is cached as a processing product for subsequent reductions from the same observing run and blue-chop setting.

Each accepted ThAr exposure is then extracted and wavelength calibrated. Messages from different frames can be interleaved during parallel processing. Inspect warnings and the number of identified and retained ThAr lines for every exposure. The original tutorial normally produced more than 900 identified lines and more than 800 retained lines per ThAr frame; a substantial shortfall or an outlying solution should be investigated.

### A.4 Stellar-frame reduction

The 15 accepted stellar exposures are bias/background corrected, extracted, flat-fielded and wavelength calibrated. Each science exposure produces an extracted `J*.mat` frame and an associated `J*_prc.mat` processing record. In the 2.1 layout these are placed under `Reduced_Data/hd_48501/reducedframes/` and `Reduced_Data/hd_48501/reducedframes/processing_files/`, respectively.

The wavelength arrays in `J*.mat` remain in the observatory frame. MEGARA records the calculated barycentric correction with the reduced frame, then applies it when constructing the post-reduction products if `options.apply_barycentric_correction=true`. This distinction is important when comparing a reduced frame directly with `final_data`.

## A.5 Post-reduction processing

With the settings above, MEGARA performs the following operations:

1. fits the continua, using synthetic guidance and SUPPNet in automatic mode;
2. applies the conservative post-reduction cosmic-ray cleanup;
3. merges the echelle orders across the standard 3869–6866 Å range;
4. applies the saved barycentric correction to the merged spectra;
5. calculates regularised LSD profiles; and
6. measures radial velocities and line widths from those profiles.

## A.6 Alternative: create a manual continuum fit

Manual fitting is the precision option when an automatic or template fit is not satisfactory. It constructs one continuum for each echelle order from points selected by the user. To use it for the tutorial, change the continuum settings before starting the post-reduction stage:

```
options.teff = 7000;
options.logg = 4.0;
options.vsinl = 44;
options.chunked_post_reduction = false;
options.manual_continuum_fit = true;
options.template_continuum_fit = false;
options.automatic_continuum_fit = false;
options.suppress_figures = false;
```

The temperature and gravity are the suggested tutorial values given above. The 44 km s<sup>-1</sup> broadening is a practical starting value. Note that manual fitting cannot be created during chunked post-reduction. Back up any existing `supplementary_data/datacf_hd_48501.mat` before this run, because creating a new manual fit overwrites it.

MEGARA opens MATLAB Figure 100 for the first order and prints four commands in the command window:

```
1 - add point
2 - delete point
3 - move point
4 - exit
```

The upper panel initially shows the representative observed order in blue and the observed intensity divided by the synthetic intensity in green (Figure 7). The synthetic spectrum is a guide to the locations and approximate widths of stellar lines; it is not expected to match every line depth or wavelength exactly.

Choose command 1, then click a continuum location in Figure 100. The chosen anchor is shown as a red cross. Repeat until at least seven anchors have been added. Once seven points are present, MEGARA draws the spline continuum in the upper panel and displays the normalised result with the red synthetic guide in the lower panel. Command 2 deletes the selected nearest anchor. Command 3 first selects the nearest anchor, then moves it to the position of the next click.

Use the following checks while placing anchors:

- Select the upper continuum envelope rather than points inside absorption lines. Line shoulders and regions where the blue and green traces approach one another can provide useful guidance.
- Anchor both ends of the order, but avoid unreliable edge pixels.
- Keep the spline smooth. Closely spaced points can generate spikes or oscillations.

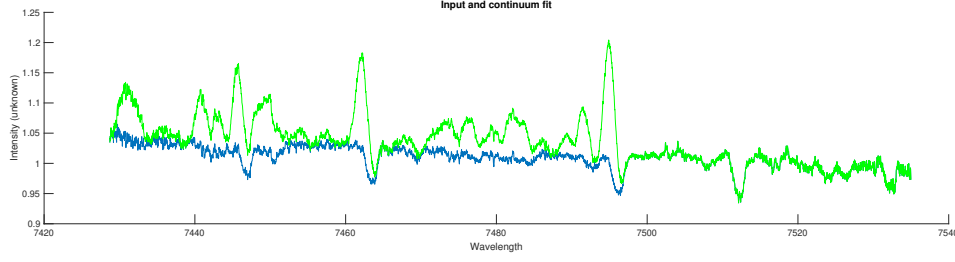


Figure 7: The initial manual-continuum display for one tutorial order. The blue trace is the representative observed spectrum and the green trace is the observed spectrum divided by the synthetic guide. No continuum anchors have yet been selected.

- Treat broad hydrogen lines carefully. The continuum should bridge the absorption smoothly rather than follow the line core.
- Seven anchors are the minimum. Roughly 7–12 well-spaced points are usually preferable to many points that begin to fit spectral structure.
- Judge the normalised lower panel as well as the upper fit. A mismatch in individual synthetic line depths is less important than a stable continuum near unity outside stellar lines.

Figure 8 shows a smooth fit with distributed anchors. Figure 9 gives a second example in a more line-rich order.

When the current order is satisfactory, enter command 4 to accept it and continue to the next order. Repeat until all requested orders have been completed. MEGARA then saves the fit as `supplementary_data/datacf_hd_48501.mat` and continues with order normalisation and merging. If the fitting session is interrupted before it finishes, the new fit may not be saved.

On later runs, set all three continuum-creation options to false while leaving `options.apply_continuum_fit=true`; MEGARA will load the saved manual fit. Set `options.manual_continuum_fit=true` only when you intend to replace it.

## A.7 Products to inspect

The main target directory should contain `final_data_hd_48501.mat` and `regularised_lsd_profile_hd_48501.mat`. Reduction and processing records are kept under `reducedframes`; continuum, RV/line-width and other supporting records are kept under `supplementary_data`. See the main output-files section for the complete layout.

Confirm that all 15 spectra expected from the tutorial are represented, that their observation times are distinct and ordered sensibly, and that the merged intensities and LSD profiles are finite. Review the continuum diagnostics for discontinuities or failed fits. The saved `vbroad` includes instrumental broadening; the saved `vsini` has the calibrated instrumental contribution removed, but neither quantity has a macroturbulence correction.

The remaining figures compare the original MEGARA 1.6 tutorial output with the same

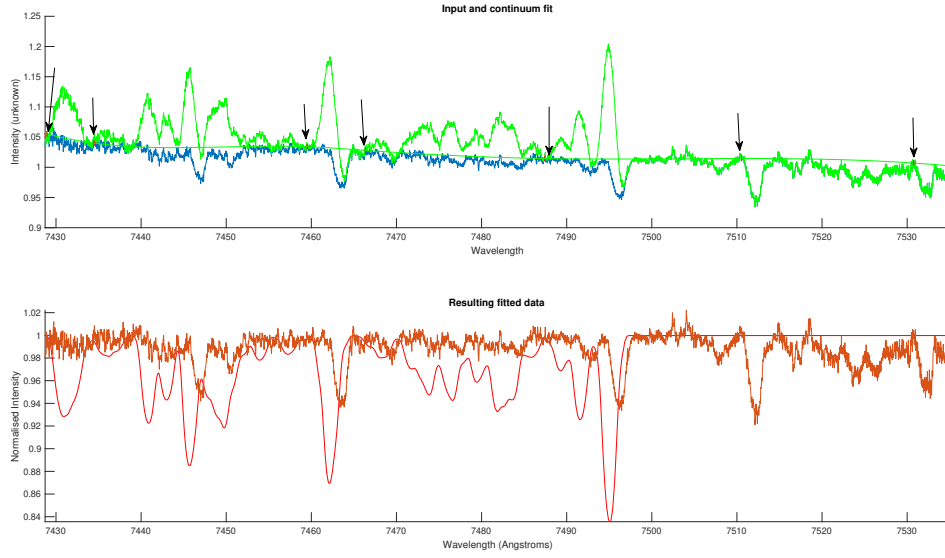


Figure 8: A completed tutorial-order fit. The selected anchors are indicated by arrows in this annotated example; the smooth green curve is the fitted continuum. The lower panel compares the normalised observed spectrum with the red synthetic guide.

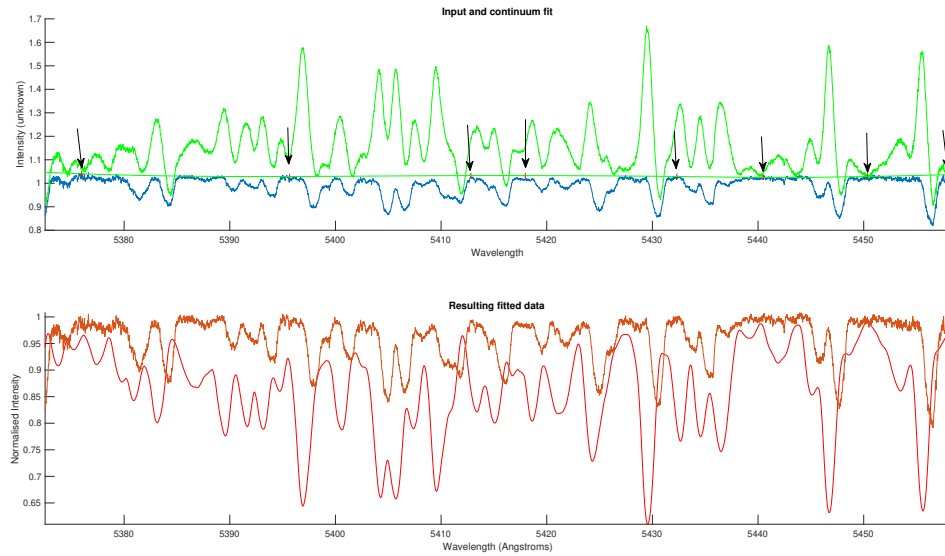
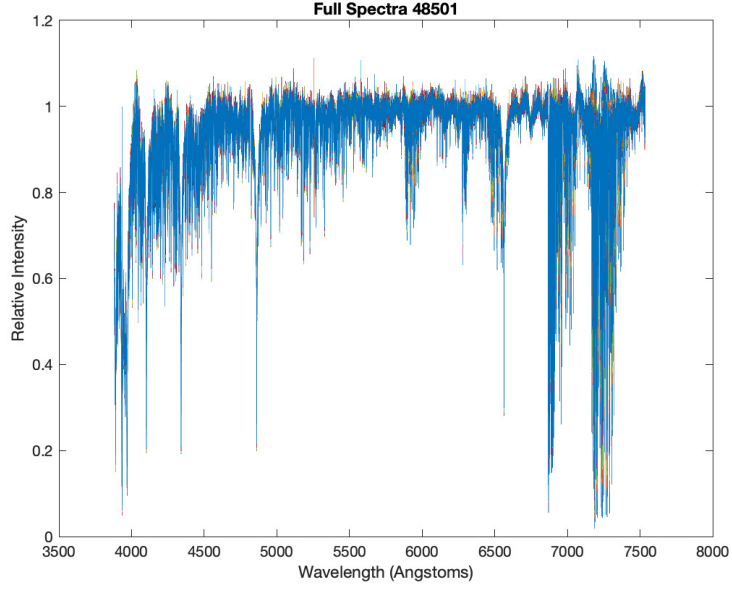
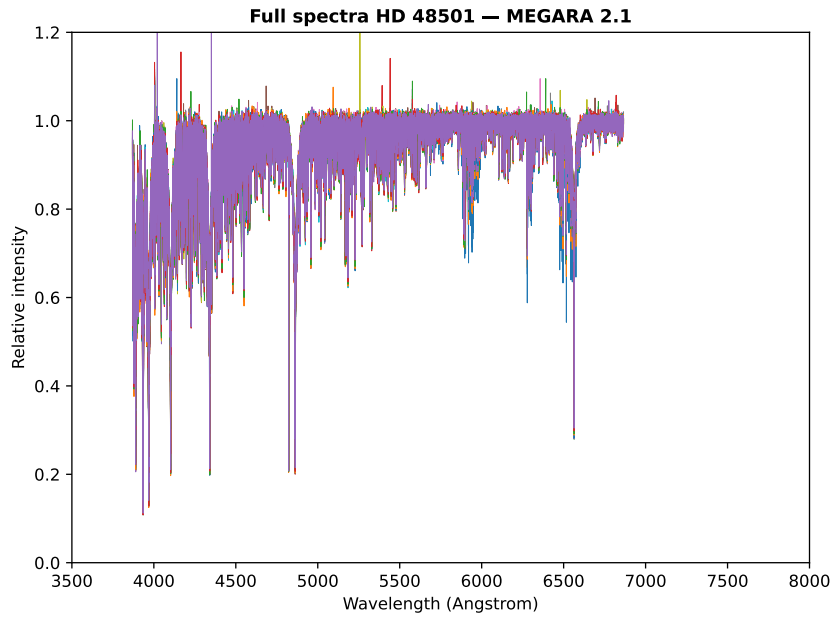


Figure 9: A second manual fit in a more line-rich region. The anchors follow the upper envelope without tracking individual absorption lines, and the lower panel provides the resulting normalised spectrum.

HD 48501 observations reduced using MEGARA 2.1. In particular, MEGARA 2.1 replaces the CCF profile with a regularised LSD profile, uses a different standard wavelength range, and reports both `vbroad` and the instrumentally corrected `vsini`.

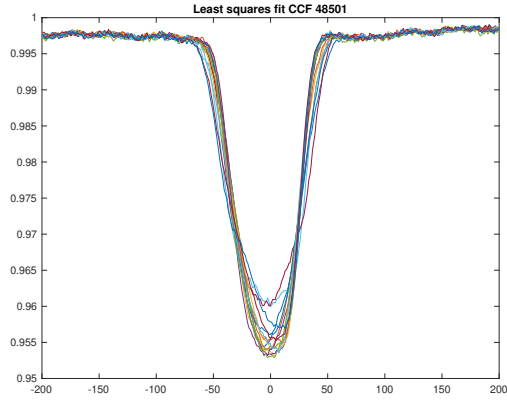


(a) MEGARA 1.6

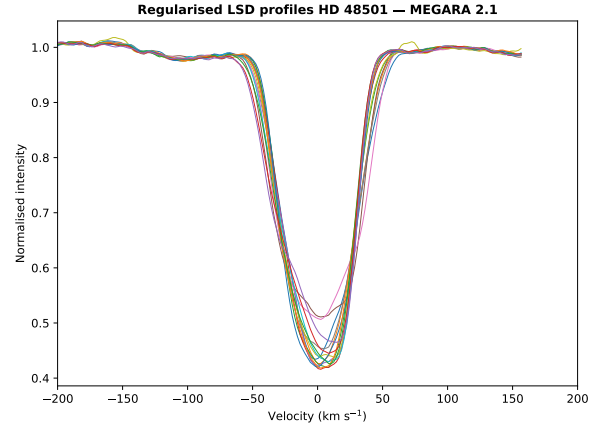


(b) MEGARA 2.1

Figure 10: Full order-merged spectra of HD 48501 from the MEGARA 1.6 and 2.1 tutorial reductions. Each colour represents a different observation. Inspect these plots for continuum shape, invalid regions and order-merging discontinuities.

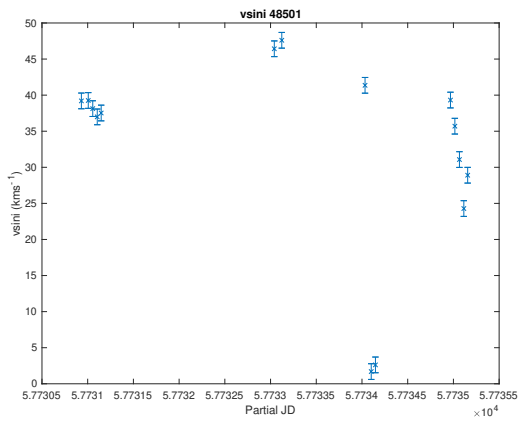


(a) MEGARA 1.6 CCF profiles

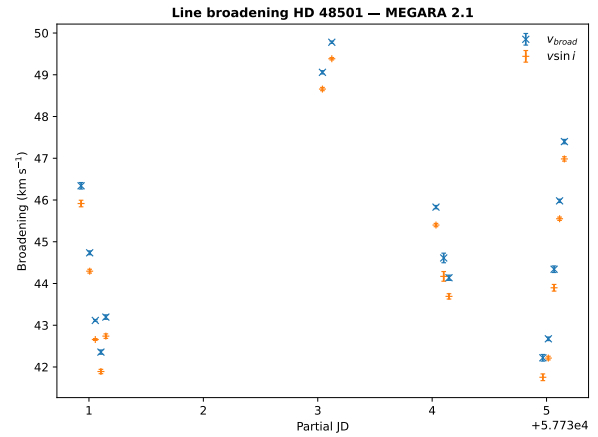


(b) MEGARA 2.1 regularised LSD profiles

Figure 11: Mean line profiles for the HD 48501 tutorial observations. MEGARA 1.6 used least-squares-fit CCF profiles; MEGARA 2.1 uses regularised LSD profiles shown after removal of the saved systemic velocity. Their depths are therefore not directly comparable.

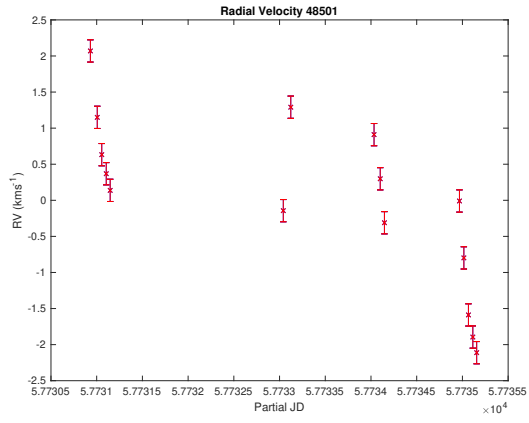


(a) MEGARA 1.6 vsini

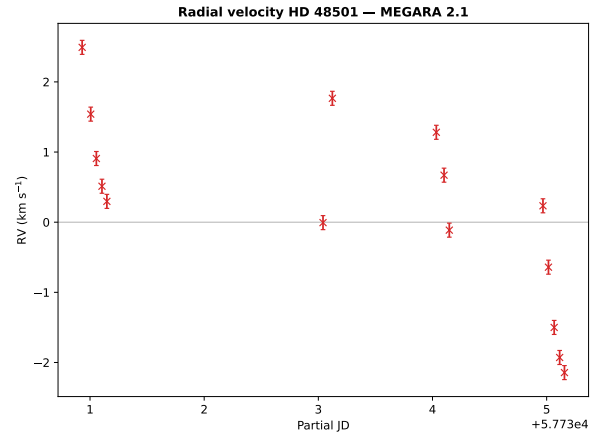


(b) MEGARA 2.1 vbroad and vsini

Figure 12: Line-broadening measurements for the HD 48501 tutorial. The two very low legacy values illustrate why individual epochs and their profiles should be reviewed. The MEGARA 1.6 quantity included instrumental broadening and corresponds to MEGARA 2.1 vbroad; the 2.1 vsini has the instrumental contribution removed but is not corrected for macroturbulence.



(a) MEGARA 1.6



(b) MEGARA 2.1

Figure 13: Relative radial-velocity measurements for the HD 48501 tutorial. MEGARA 2.1 RVs are measured from the first moments of the regularised LSD profiles, so they need not reproduce the MEGARA 1.6 CCF measurements.

For a repeat post-reduction test without extracting the raw frames again, set `options.skip_reduction=true`, retain `options.skip_post_reduction=false`, and choose the overwrite settings for the products that you intend to rebuild. Set `options.star_census=false` once the raw-data inventory is unchanged.

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